Similar Shapes: Area & Volume

Videos 293a & 293b on Corbettmaths

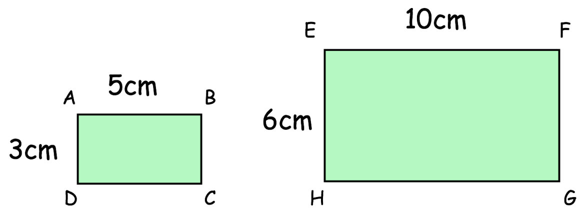
Examples

Workout 1

Click here | Scan here — Similar Shapes: Area

Click here | Scan here — Similar Shapes: Volume

Question 1: Rectangle EFGH is an enlargement of rectangle ABCD.

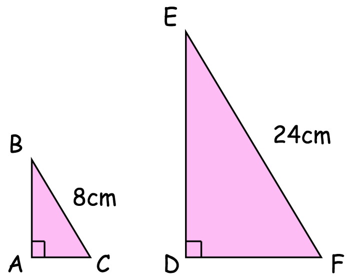


(a) What is the scale factor of enlargement?

(b) How many times larger is the area of EFGH than ABCD?

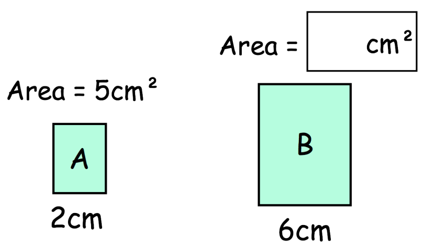
Question 2: Triangles ABC and DEF are similar.

How many times larger is the area of ABC than ABCD?

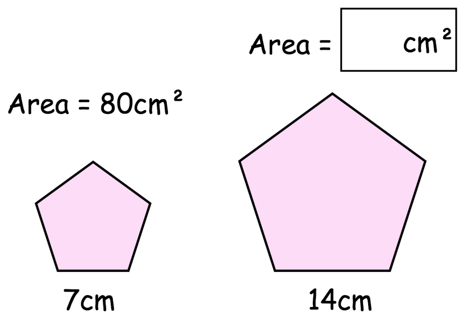


Question 3: Each pair of shapes below are similar.
Find the missing areas.

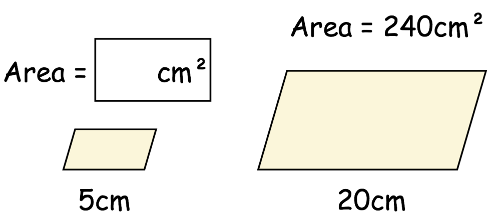
(a)



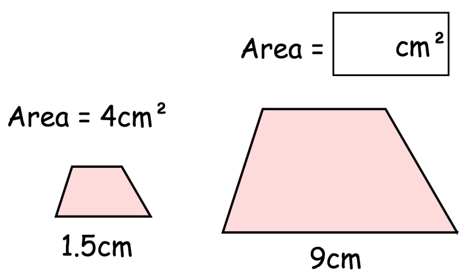
(b)



(c)



(d)



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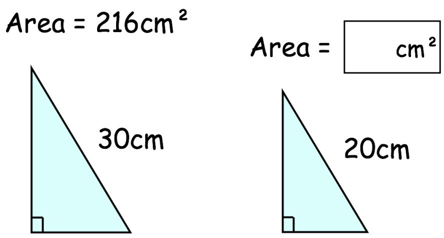
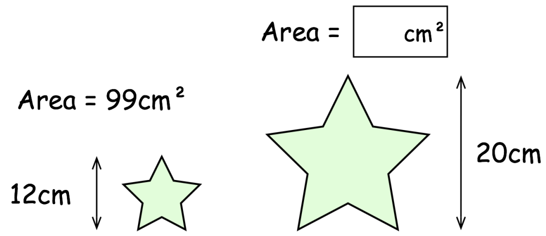
(e)

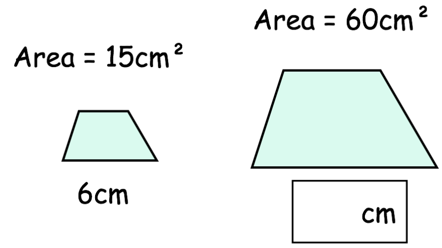


(f)

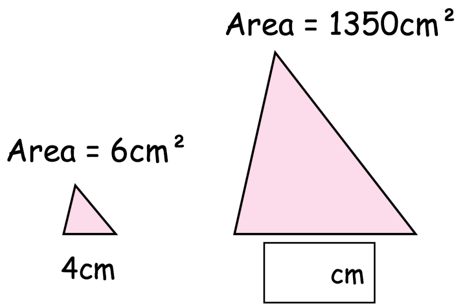


Question 4: Each pair of shapes below are similar.
Find the missing lengths.

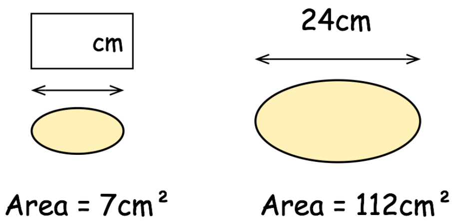
(a)



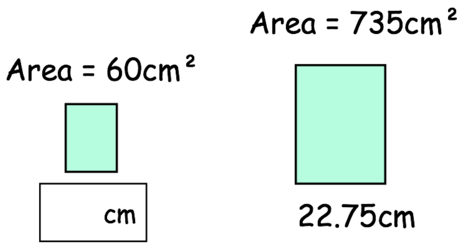
(b)



(c)

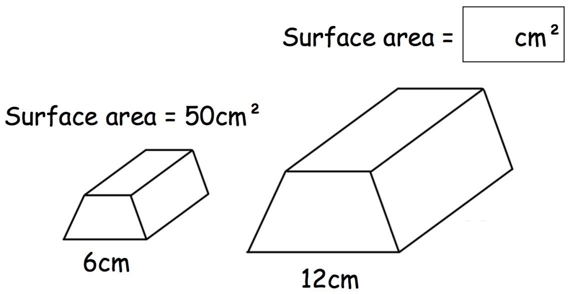


(d)

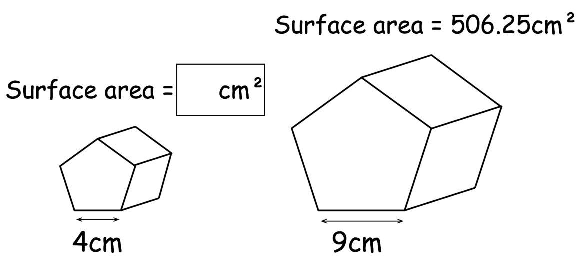


Question 5: The solid shapes below are mathematically similar.
Find the missing surface areas.

(a)

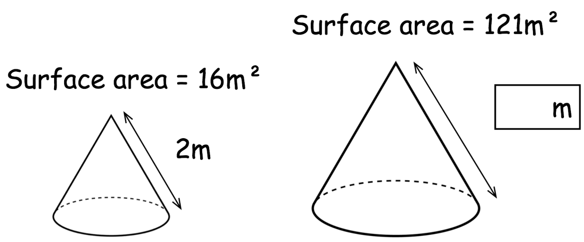


(b)

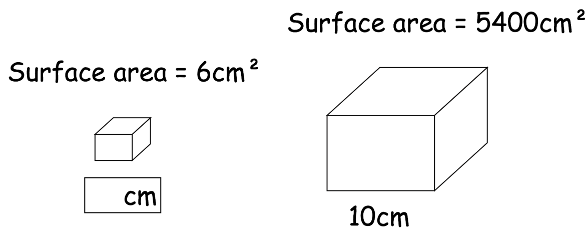


Question 6: The solid shapes below are mathematically similar.
Find the missing lengths.

(a)



(b)



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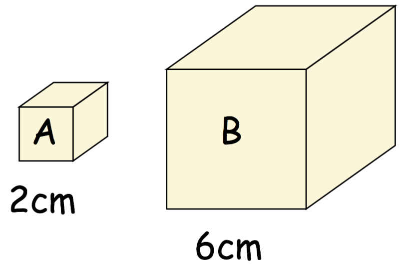
Workout 2

Question 1: Cube B is an enlargement of cube A.

(a) What is the scale factor of enlargement?

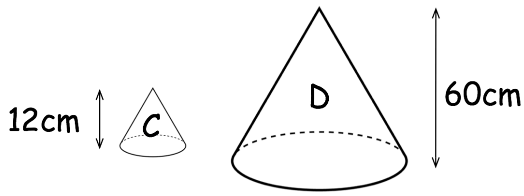
(b) How many times larger is the surface area of B than A?

(c) How many times larger is the volume of cube B than A?



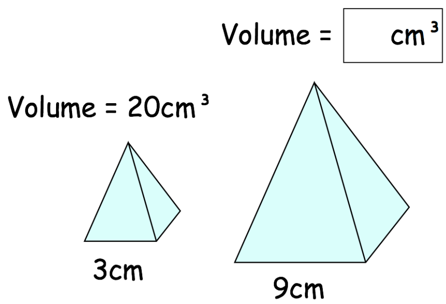
Question 2: Cones C and D are similar.

How many times larger is the volume of D than C?

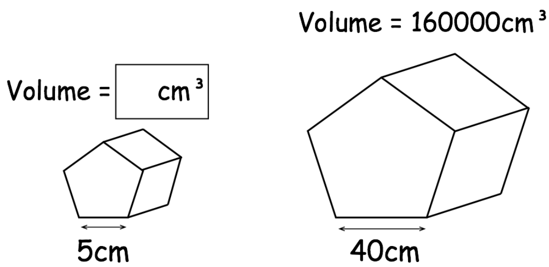


Question 3: The solids below are mathematically similar.
Find the missing volumes.

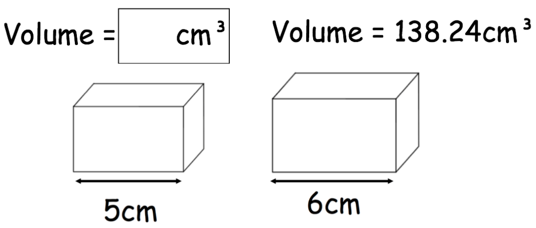
(a)



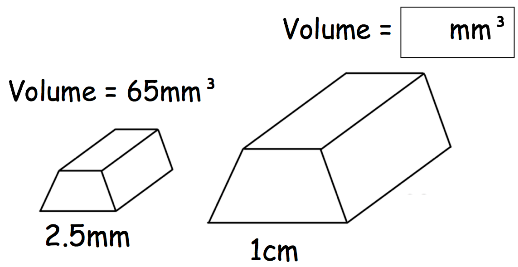
(b)



(c)

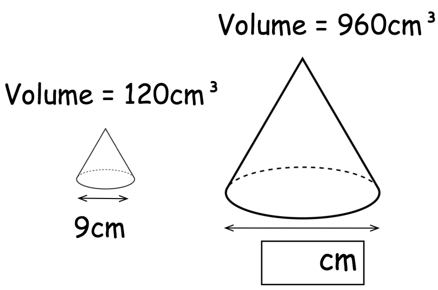


(d)

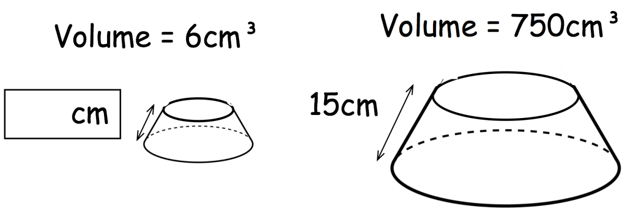


Question 4: The solid shapes below are mathematically similar.
Find the missing lengths.

(a)



(b)



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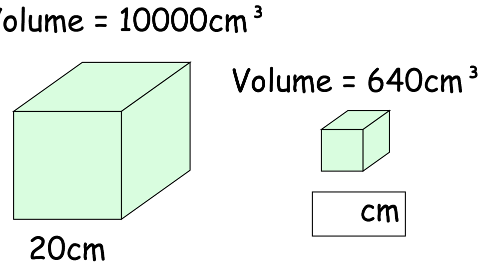
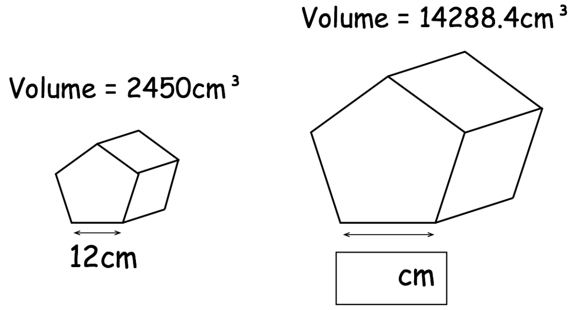
(c)

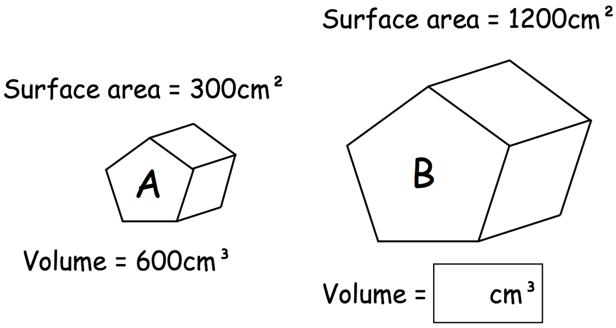


(d)

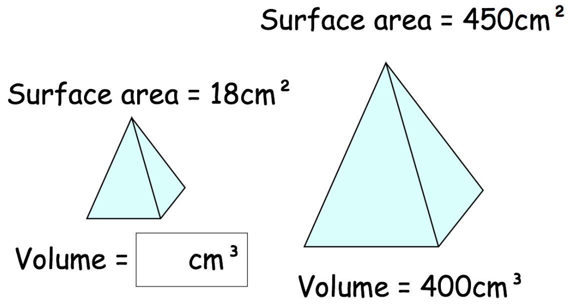


Question 5: The solid shapes below are similar.
Find the missing volumes.

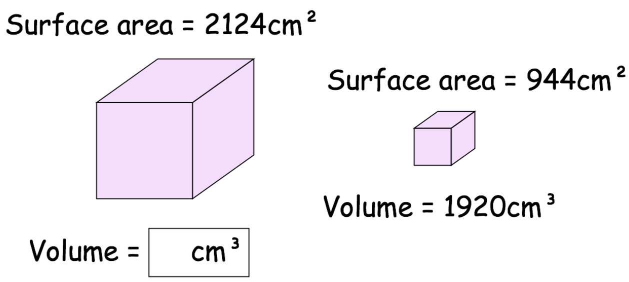
(a)



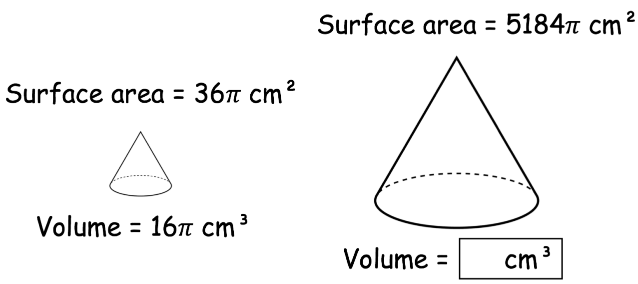
(b)



(c)

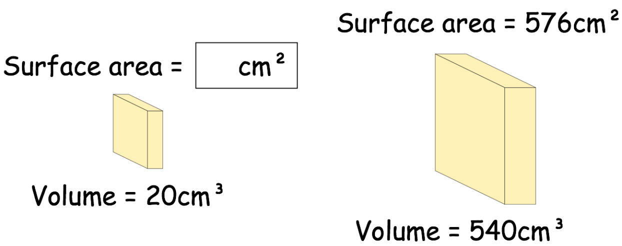


(d)

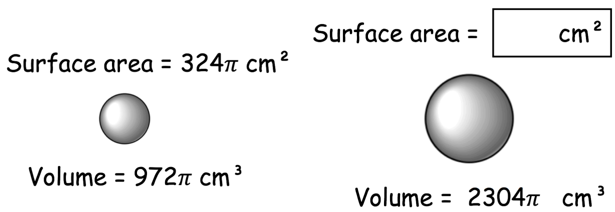


Question 6: The solid shapes below are similar.
Find the missing surface areas.

(a)



(b)



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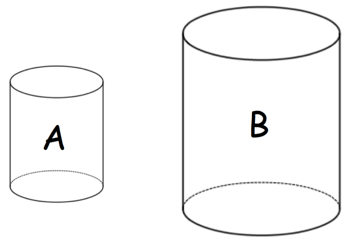
Workout 3

Question 1: A and B are mathematically similar.

The height of A : the height of B = 3 : 5

(a) Find the surface area of A : the surface area of B

(b) Find the volume of A : the volume of B

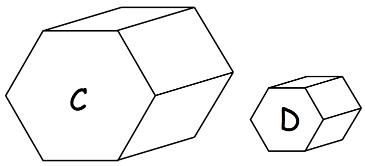


Question 2: Solids C and D are similar.

The length of C : the length of D = 9 : 2

(a) Find the surface area of C : the surface area of D

(b) Find the volume of C : the volume of D

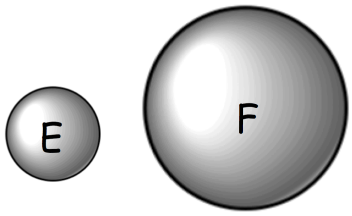


Question 3: Shown are spheres E and F.

The surface area of E : the surface area of F = 4 : 49

(a) Find the diameter of E : the diameter of F

(b) Find the volume of E : the volume of F

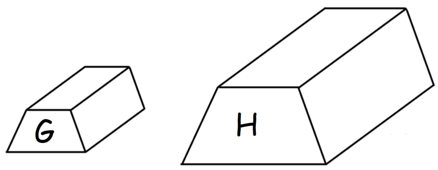


Question 4: Shown are similar solids G and H.

The volume of G : the volume of H = 27 : 1000

(a) Find the height of G : the height of H

(b) Find the surface area of G : the surface area of H



Question 5: The surface areas of two similar shapes are in the ratio $25 : 81$
The length of the smaller shape is 30cm.

Work out the length of the larger shape.

Question 6: The volumes of two similar shapes are in the ratio $1000 : 27$
The surface area of the larger shape is $250cm^2$

Work out the surface area of the smaller shape.

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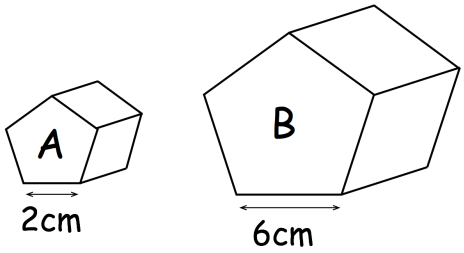
Apply

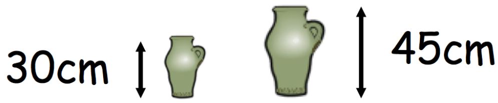
Question 1: Myles says that the surface area of B is three times larger than the surface area of A.

Explain why Myles is incorrect.



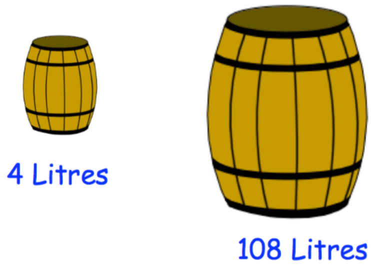
Question 2: Two vases are similar.
The smaller vase holds 800ml.

How much does the larger vase hold?



Question 3: Two barrels are mathematically similar.
It takes 90ml of paint to paint the smaller barrel.

How much paint is needed for the larger barrel?



Question 4: A large bottle of cola is 16cm tall.
A small bottle of cola is 12cm tall.
The bottles are mathematically similar.
Fernando claims the small bottle contains three-quarters of the amount of cola than the large bottle.

Show Fernando is wrong.

Question 5: Nina bought a toy that grows when placed in water.
Before placing the toy in water, it was 4cm tall.
After placing the toy in water it grew to a similarly shaped toy that was 11cm tall.

Is the claim that it will “grow 20 times larger” reasonable?

Question 6: Alec makes two solid mathematically similar models of the Titanic.

The smaller model had a length of 10cm
The larger model had a length of 45cm
The mass of the larger model is 7.29kg

Work out the mass of the smaller model.

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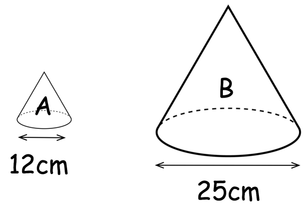
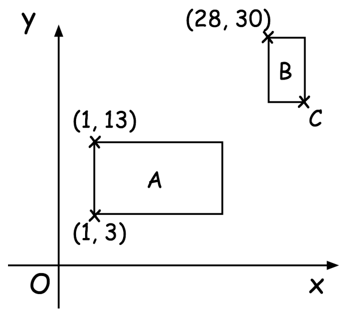
Question 7: Below are two similar cones.



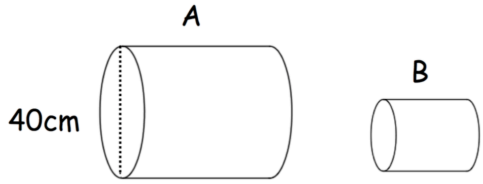
Anna says the volume of A is approximately 11% of the volume of B.
Is Anna correct? Explain your answer.

Question 8: Rectangles A and B are similar.
The area of rectangle A is 240
The area of rectangle B is 15.

Find the coordinates of the point C.



Question 9: Cylinders A and B are similar.



$$\text{Volume B} \times \frac{512}{27} = \text{Volume A}$$

Find the radius of cylinder B.

Question 10: A, B and C are similar.

The volume of A is 729cm^3 and the volume of B is 64cm^3.

The surface area of B is 25cm^2 and the surface area of C is 121cm^2.

Find the ratio length of A : length of B : length of C

Question 11: Shapes A, B and C are similar.
The height of shape A is 8cm.
The height of shape C is 4cm.
The ratio of the surface area of shape B to the surface area of shape C is 25:9

Work out the ratio of the volume of shape A to shape B.

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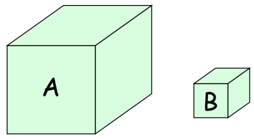
Question 12: Two solid toys, C and D, are similar.
The volume of toy C is $40cm^3$

The surface area of C : surface area of D = 2 : 9

Work out the volume of toy D.

Question 13: Washing powder is sold in two different sizes, a large box A and a smaller box B.
Cuboid boxes A and B are similar.

Surface area of A : Surface area of B = 81 : 4



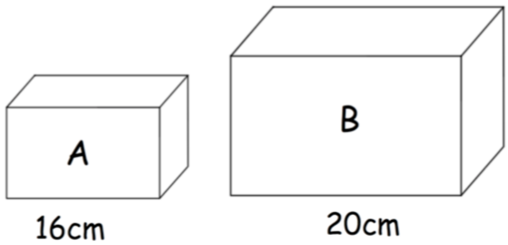
How many smaller boxes, B, can be completely filled using the contents of a full box A?

Question 14: Pyramid A and pyramid B are similar.

The surface area of B is 42.25 times larger than the surface area of A.

Find the ratio of the volume of A to the volume of B.

Question 15: Cuboids A and B are similar but made from different materials.
Both cuboids are placed on a table.



The pressure on the table due to cuboid A is 3.5 newtons/cm^2
Cuboid A exerts a force of 420N on the table.

The pressure on the table due to cuboid B is 4 times larger than cuboid A.

Work out the force exerted by cuboid B on the table.

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Question 16: Ornament A and B are mathematically similar.
They are solid and both made from copper and zinc in the ratio 3:2

Ornament A has a height of 5cm and volume of 30cm^3
Ornament B has a height of 18cm.

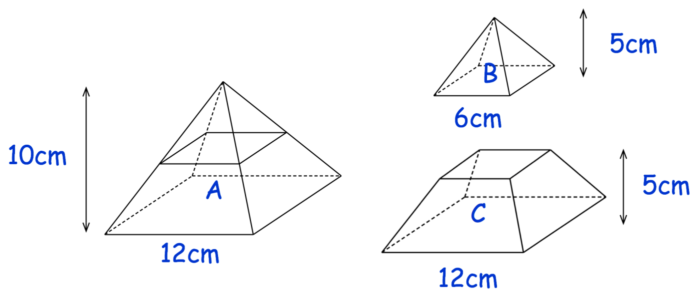
The density of copper is 8.96g/cm^3
The density of zinc is 7.13g/cm^3

Work out the difference in mass between ornament A and ornament B.

Question 17: Cylinders A and B are similar.
The height of A is 6cm.
The volume of A is 240cm^3 to 2 significant figures.
The height of B is 15cm and the volume of B is y.

Work out the error interval of y.

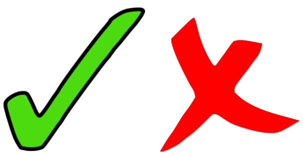
Question 18: The square based pyramid A is divided into Pyramid B and Frustum C.



(a) Express the volume of Pyramid B as a fraction of the volume of Pyramid A.

(b) Express the volume of Frustum C as a fraction of the volume of Pyramid A.

Answers



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